

Population Aging, Tasks, and Regional Growth: A Quantitative Analysis of Japan*

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Abstract

Population aging changes not only the size of the labor force but also its task composition. We study this channel in Japan, where population aging is especially severe. We develop a quantitative spatial general equilibrium model with task-based labor supply, endogenous agglomeration, and Eaton-Kortum sourcing across 47 prefectures and the rest of the world. Age cohorts differ in task capabilities: younger workers have a comparative advantage in non-routine cognitive tasks, while older workers are concentrated in routine and manual tasks. As the young workforce contracts, regions with high baseline exposure to non-routine cognitive production gain through the technical-access channel. Metropolitan cores sustain output and retain young labor through cognitive-labor concentration, while peripheral regions face weaker productivity gains and larger cost increases. Welfare losses are pervasive but spatially uneven: young-worker retention compresses local scarcity premia in cities, while peripheral regions experience amenity losses and stronger cost pass-through. Senior workers face the sharpest losses everywhere, since their labor supply expands without an offsetting amenity mechanism. An automation extension reduces routine-task demand, accelerating reallocation toward non-routine work and raising market exit among the young.

Keywords: aging economy, Eaton-Kortum model, task-based approach, agglomeration, Japan

JEL Classification: J14, J24, R12, R13

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1 Introduction

Japan’s labor force has been shrinking since the mid-2000s, and the age composition of the workforce is shifting sharply toward older cohorts. The share of workers aged 65 and over has risen steadily and, under official projections, will continue to rise through mid-century. Population aging of this scale and duration forces structural change: the workers being lost are not a representative cross-section of the labor force, so sectors and regions do not simply contract in proportion. How the burden falls across space depends on what those workers were doing. We argue that understanding the spatial consequences of Japan’s demographic transition requires attention to the task content of local production, a connection between demography and economic geography that has received growing attention in the macroeconomic literature ([Cai and Stoyanov 2016](#); [Herrendorf et al. 2014](#); [Hopenhayn et al. 2022](#)).

A substantial literature documents that physical and cognitive capacities evolve over the life cycle, generating distinct task supply profiles across age groups ([Feyrer 2007](#); [Muto et al. 2016](#); [Gu and Stoyanov 2019](#); [Deming and Noray 2020](#); [Maestas et al. 2023](#)). Older workers tend to specialize in experience-based and routine tasks; younger workers hold a comparative advantage in tasks requiring fluid intelligence and physical stamina. [Autor and Dorn \(2009\)](#) show that occupations exhibit systematic age structures, so that demographic shifts directly force a reallocation of task-specific labor. These supply-side shifts interact with the task intensity of sectoral production ([Autor et al. 2003](#); [Autor and Dorn 2013](#)). While demand-side channels, such as rising expenditure on elder care, are well documented ([Cravino et al. 2022](#); [Badilla Maroto et al. 2026](#)), we focus on the supply side and ask how the scarcity of younger workers alters relative factor prices and the spatial allocation of labor ([Acemoglu and Restrepo 2022](#)).

Standard labor market studies of demographic change ask whether elderly employment crowds out younger workers ([Gruber et al. 2009](#); [Bianchi et al. 2023](#)). We depart from that partial equilibrium view. Tasks are not perfect substitutes across age groups, and most sectors combine cognitive and manual inputs in production. When the young cohort shrinks, the resulting scarcity is not spread evenly. It concentrates in sectors and regions that depend most on cognitive labor, and it propagates through spatial input-output linkages in ways that a single-market framework cannot capture ([Mortensen and Pissarides 1999](#)). The central question is therefore not whether older workers displace younger ones, but how the localized scarcity of cognitive labor reshapes comparative advantage and spatial economic structure.

The central mechanism we identify is the spatial sorting of cognitive labor. Metropolitan areas already attract a disproportionate share of high-skilled workers in normal times ([Chen et al., 2024](#)). We endogenize this tendency by modeling the local Eaton-Kortum technology scale as a function of the regional concentration of non-routine cognitive workers, capturing the localized agglomeration economies documented in the urban economics literature ([Moretti,](#)

2004). As the young workforce contracts nationally, non-routine cognitive workers grow scarce and concentrate further in cities like Tokyo. This concentration raises local technology through agglomeration economies, and the gains diffuse through bilateral sourcing links. Urban cores can therefore sustain output growth even as the aggregate workforce shrinks, because the sorting of scarce cognitive talent intensifies the productivity advantages that drive agglomeration in the first place.

A structural decomposition of amenity-adjusted market payoffs reveals a more complicated picture for metropolitan areas. The scarcity of young cognitive workers generates nominal wage premiums, but these premiums are partly arbitrated away in regions that retain young market labor. Tokyo sustains output and has the smallest decline in young-market amenities, but the inflow of young workers compresses the local scarcity premium. Some non-metropolitan regions instead experience larger percentage payoff gains for the remaining young cognitive workers, even as their output falls. Senior workers bear the largest losses because the expansion of older labor supply lowers their own wage component and because cost increases reduce purchasing power.

Peripheral regions follow a different path. They do not benefit from the endogenous productivity gains that accrue to metropolitan cores, but their task structure is more compatible with an older workforce. Routine and manual tasks dominate production in many peripheral prefectures, and these tasks do not require the fluid cognitive capabilities that older workers lack. This compatibility allows peripheral regions to absorb an expanding elderly labor supply without severe wage collapse, although weaker technology growth and input-cost exposure still reduce real income. Regional vulnerability to demographic change is therefore determined not by population size alone, but by whether local task demand aligns with the task supply that an aging workforce can provide.

Together, these mechanisms imply that population aging is a force for spatial divergence, not convergence. Metropolitan areas grow more productive and retain young market labor, while many peripheral areas miss the largest technology gains and face less favorable cost conditions. The incidence across workers does not follow the output ranking one-for-one: young cognitive workers can receive larger percentage payoff gains in shrinking regions where they become scarce, while elderly workers face losses nearly everywhere.

This paper contributes to the literature in three ways. First, it integrates the task-based approach to labor markets (Acemoglu and Autor 2011) into a quantitative spatial general equilibrium framework with endogenous agglomeration economies and Eaton-Kortum trade (Redding and Rossi-Hansberg 2017; Eaton and Kortum 2002), establishing task structure as a determinant of regional comparative advantage under demographic change. While both literatures are well developed separately, their intersection has received limited attention, and we show that the combination generates predictions about spatial inequality that neither framework produces in isolation. Second, we derive a structural decomposition of amenity-

adjusted market payoffs into own scarcity, endogenous amenities, technical access, sourcing reallocation, and cost channels. This decomposition shows why production incidence and worker incidence need not coincide: young labor retention supports output in metropolitan areas, while local scarcity can generate larger percentage payoff gains for young cognitive workers who remain in shrinking regions. Third, we use Japan as a quantitative laboratory and discipline the model with prefecture-level data, regional productivity regressions, and official demographic forecasts. The calibrated agglomeration elasticity for non-routine cognitive labor implies agglomeration returns that are substantially larger than canonical estimates based on college shares, a difference we trace to the precision of the task-based measure.

The remainder of the paper is structured as follows: Section 2 presents motivating evidence. Section 3 introduces the model framework. Section 4 quantifies the model and Section 5 performs counterfactual analysis. Finally, Section 6 concludes.

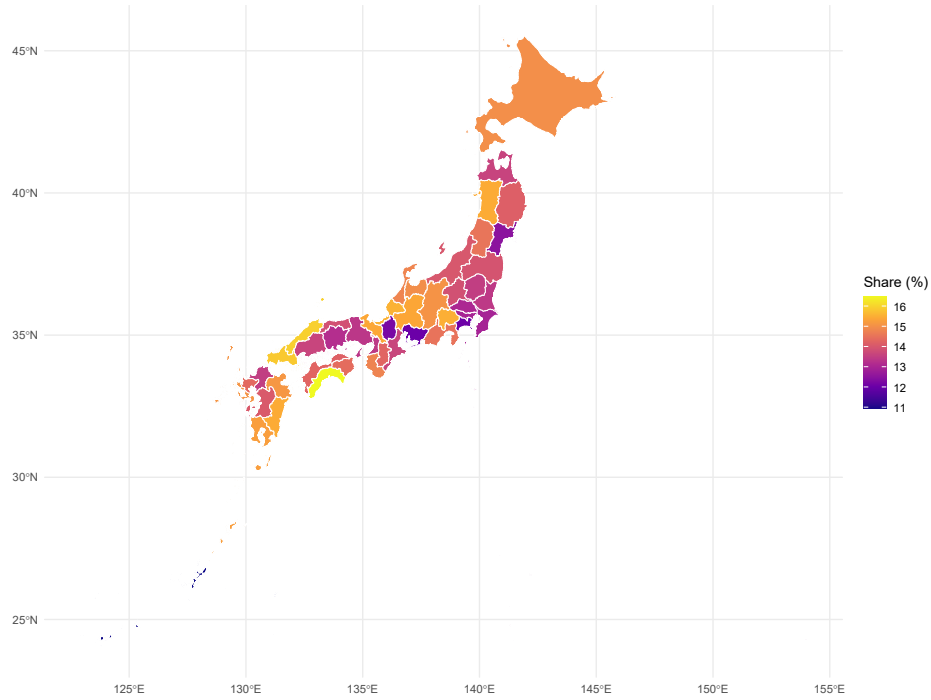
2 Evidence

This section establishes three empirical patterns that motivate the model. First, elderly workers are geographically concentrated in peripheral prefectures rather than in major cities, so the demographic shock is not spatially uniform. Second, elderly workers supply task inputs differently from younger cohorts: they are disproportionately concentrated in routine and manual occupations, while non-routine cognitive work is dominated by younger workers. Third, regional innovation is positively associated with the local concentration of non-routine cognitive workers and is uncorrelated with the concentration of other task types, providing direct support for the endogenous productivity specification. All evidence draws from the *2022 Japanese Employment Status Survey* unless noted otherwise.

At the regional level, the share of elderly workers exhibits significant geographic disparities. As shown in Figure 2, the highest proportions are concentrated in peripheral and rural regions, led by Kochi (16.5%), Shimane (15.9%), and Yamaguchi (15.8%) prefectures. This pattern reflects the ongoing out-migration of younger demographics, leaving local labor markets highly dependent on older cohorts. Conversely, major economic hubs maintain relatively lower concentrations, with Tokyo, Osaka, and Aichi reporting elderly worker shares of 13.0%, 13.4%, and 12.0% respectively, while the national minimum is observed in Okinawa (10.9%). These findings reinforce that population aging is not spatially uniform; rather, peripheral prefectures bear a disproportionate burden of the demographic shock.

Further analysis reveals a notable concentration of elderly workers in specific sectors and a steady increase in their participation over time. From 2007 to 2022, the share of workers aged 65 and over increased across all broad industry categories. According to the 2022 Japanese Employment Status Survey, the proportion of elderly workers in the manufacturing sector rose from 5.32% to 8.78%, and the service sector experienced an increase from 4.60% to 9.26%.

(a) Share of Workers Aged 65+ in Each Region in 2022



(b) Prefectures of Japan

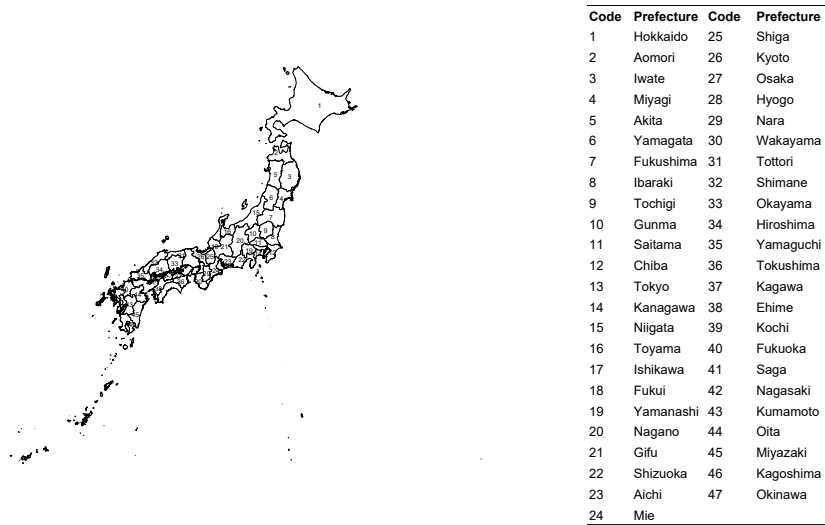


Figure 1

Figure 2: Distribution of Workers Aged 65+

Table 1: Labor Distribution by Task Categories and Age Segments (%)

Task	Young (15–34)	Middle (35–64)	Senior (65+)
Non-routine non-manual	20.33	76.51	3.16
Non-routine manual	28.30	67.76	3.94
Routine non-manual	27.61	65.30	7.09
Routine manual	22.31	70.51	7.17

To characterize the age structure of task supply directly, we classify occupations into four task categories following [Doi and Suzuki \(2025a\)](#), integrating occupational microdata from the 2022 Japanese Employment Status Survey with task intensity metrics from the Occupational Information Network (ONET). We employ a crosswalk procedure through the International Standard Classification of Occupations to map the US based ONET descriptors to the corresponding Japanese occupational codes.

For each occupation, we aggregate task variables into four core dimensions. Non-routine non-manual tasks encompass abstract problem solving and data analysis; we refer to this category interchangeably as non-routine cognitive tasks throughout the paper. Routine non-manual tasks involve repetitive cognitive activities including record keeping and calculating. Routine manual tasks require repetitive physical motions and machine control. Non-routine manual tasks demand physical adaptability and spatial orientation typically found in interpersonal services. We standardize these scores to have a mean of zero and a standard deviation of one across total employment, assigning each occupation to the category with its highest relative score.

Figure 3 visualizes this classification strategy by plotting specific occupations based on their routine and manual task intensities. Occupations including System Consultants and Researchers cluster in the upper left quadrant. These cognitive intensive roles require fluid intelligence and continuous skill upgrading, explaining the minimal senior participation in the Information and Communications sector. The bottom right quadrant features occupations including Building Cleaners and Taxi Drivers. These roles are characterized by repetitive physical tasks that rely on crystallized experience and reliability, directly corresponding to the high concentration of elderly workers in Other Services and Transportation sectors.

Table 1 confirms a distinct age stratification within the Japanese labor market. The prime age workforce (ages 35 to 64) accounts for 62.25% of total employment. Their dominance is most absolute in non-routine non-manual tasks (76.51%), encompassing management and professional roles that demand peak cognitive ability combined with accumulated experience.

The senior population (ages 65+) exhibits a strong comparative advantage in routine and manual roles. Despite an overall market share of 7.96%, they supply 7.17% of routine manual labor and 7.09% of routine non-manual labor. This indicates that older workers in Japan

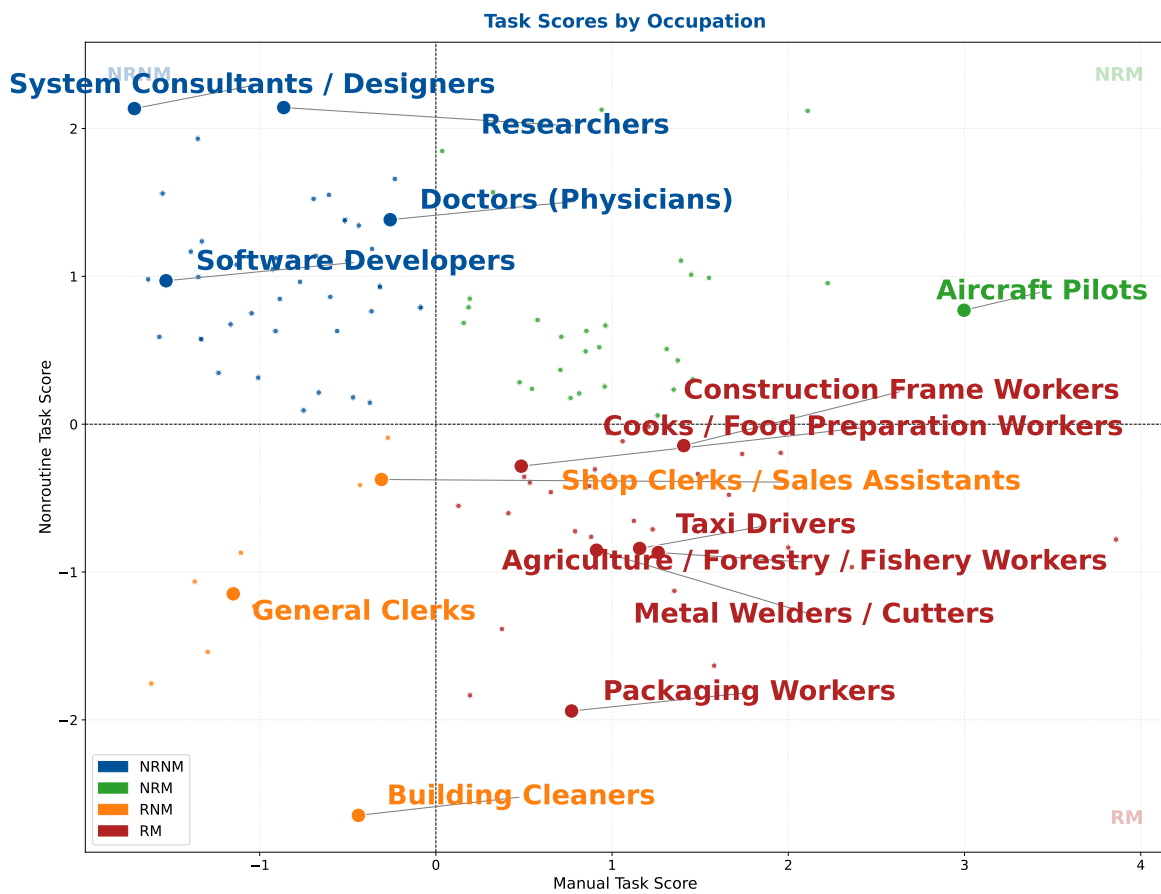


Figure 3: Task scores of occupations

concentrate in roles that rely on accumulated experience and physical repetition rather than the fluid intelligence required for non-routine cognitive tasks, where their participation drops to 3.16%. The youngest cohort (ages 15 to 34) records its highest engagement in non-routine manual tasks at 28.30%, suggesting entry level service roles serve as the primary labor market gateway.

To visualize the geographic dimension of this distribution, Figure 4 maps the spatial specialization of tasks across Japan based on relative task supply. Panel (a) shows the distribution for the total labor force. The Tokyo metropolitan area is specialized in cognitive intensive non-routine non-manual tasks, functioning as the primary hub for professional services. The industrial heartland including Aichi and Mie maintains a comparative advantage in routine manual tasks, aligning with modern manufacturing infrastructure. Peripheral regions frequently specialize in non-routine manual labor to support local service economies.

Panel (b) restricts the sample to workers aged 65 and over. In many rural prefectures, the elderly show a stronger comparative advantage in routine manual tasks than the general workforce does, indicating that older workers are the primary holders of local task specialization in peripheral regions, not a residual labor supply. Where the working-age population has contracted, local production increasingly depends on the task capabilities that older workers can provide.

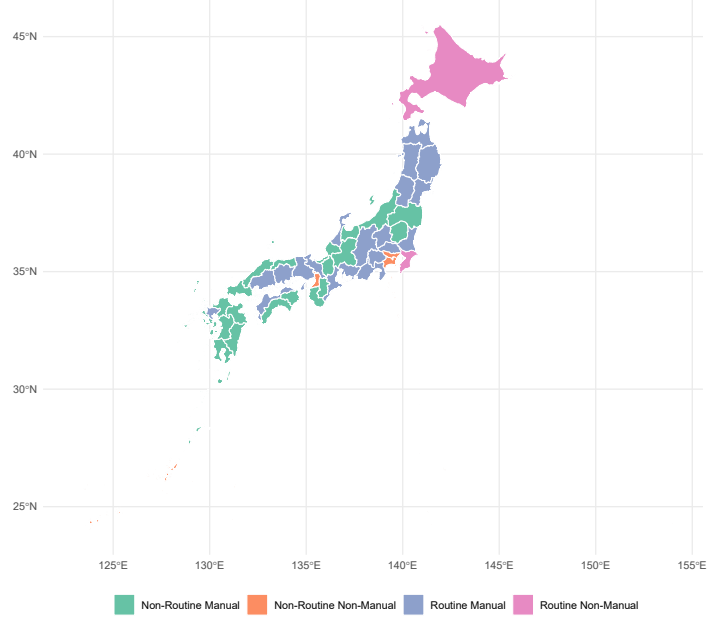
The third motivating pattern concerns the relationship between task composition and regional innovation. We measure innovation using log patent applications per worker and plot it against local employment shares for each of the four task types in Figure 5.

Non-routine cognitive labor is the only task type positively associated with regional innovation. The top left panel establishes a positive log linear relationship between the local non-routine non-manual task share and per capita patenting rates. Metropolitan hubs including Tokyo, Osaka, and Kanagawa exhibit both the densest concentration of cognitive workers and exponentially higher innovation output (Rossi-Hansberg et al., 2026). The spatial concentration of the remaining three task categories yields no technological dividend. The scatter plots for routine non-manual and non-routine manual tasks exhibit clear downward sloping trends, and the routine manual share shows zero positive correlation with local innovation. Peripheral regions dominated by routine and manual labor locate consistently at the bottom of the innovation distribution.

This cross sectional heterogeneity justifies our theoretical design. It confirms that localized increasing returns are not a generic byproduct of aggregate employment density. Regional agglomeration economies are exclusively driven by the concentration of the cognitive workforce. We therefore strictly confine the endogenous productivity mechanism in our quantitative spatial framework to the non-routine non-manual labor supply.

(a) Task with The Highest Relative Task Supply

$$\arg \max_i \left(\frac{L_n^i / \sum_{i'} L_n^{i'}}{\sum_n L_n^i / \sum_{m,i'} L_m^{i'}} \right)$$



(b) Task with The Highest Relative Task Supply of Old Workers

$$\arg \max_i \left(\frac{L_n^{i,Old} / \sum_{i'} L_n^{i',Old}}{\sum_n L_n^{i,Old} / \sum_{m,i'} L_m^{i',Old}} \right)$$

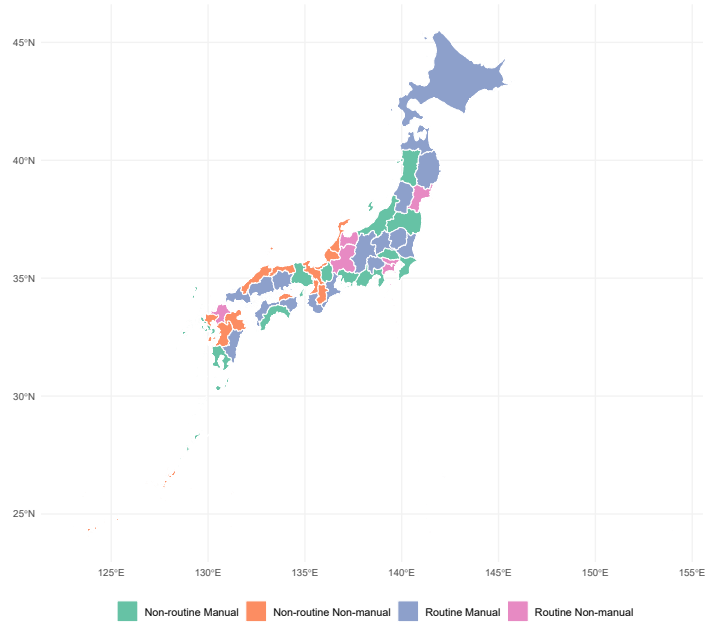


Figure 4: Spatial Specialization of Tasks in Japan

3 Quantitative Model

We construct a quantitative spatial general equilibrium model to analyze how population aging reshapes comparative advantage and regional economic structure. The framework has two central ingredients. First, labor supply is organized around tasks rather than aggregate skill levels. Age cohorts differ in their productivity across tasks, so a shift in the age composition of the workforce is equivalent to a change in the effective supply of each task type, with consequences that vary by sector and region depending on how intensively each task is used in production. Second, goods are traded across prefectures and the rest of the world through an Eaton-Kortum sourcing structure. Regional technology scales and bilateral trade costs determine sourcing shares, so demographic shocks affect amenity-adjusted market payoffs through local task costs, endogenous market amenities, and the spatial price index.

3.1 Environment

In each country, there is an ex-ante identical continuum of workers indexed by ω . We divide workers into G age groups indexed by a . In Japan, workers are initially distributed across regions. Let $\bar{L}_m^a$ denote the mass of age- a residents initially located in region m . Workers make a sequential decision. Young and middle-aged workers ($a \neq Old$) participate in the market sector only and choose a destination region n and a task i . Old workers ($a = Old$) first choose whether to retire in their origin region m or to participate in the market sector in a destination region n .

The market sector, denoted by M , includes manufacturing and services. Wages in this sector are endogenously determined by productivity and labor market clearing. The retirement sector is available only to old workers ($a = Old$). Workers in this sector do not participate in market production and do not migrate. They receive the national pension h^a , given by $h^a = \tilde{\tau}GDP/L^{Old}$, where $\tilde{\tau}$ is an exogenous transfer ratio. Retired workers spend the pension on non-traded goods and services such as healthcare and personal care, which are outside the model. Their expenditure therefore does not enter aggregate demand for traded goods. We retain the notation H for the retirement sector throughout for consistency with the solution algorithm.

If the worker chooses the market sector in destination n , she receives the age-specific market wage w_n^{Ma} and pays an origin-destination migration cost $\kappa_{nm}^a \geq 1$, where the first index denotes the destination and the second index denotes the origin. The indirect utility from choosing the market sector in destination n is

$$v_{nm}^{Ma}(\omega) = \ln \left(B_n^{Ma} \frac{w_n^{Ma}}{P_n \kappa_{nm}^a} \right) + \eta \varepsilon_{nm}^{Ma}(\omega).$$

If the worker ($a = Old$) retires in origin region m , she obtains

$$v_m^{Ha}(\omega) = \ln \left(B_m^H \frac{h^a}{P_m} \right) + \eta \varepsilon_m^{Ha}(\omega).$$

Here, B_n^{Ma} and B_m^H denote sector-specific amenities. For market workers, w_n^{Ma} is the CES task aggregate defined in (2). Since all household consumption is directed toward market goods, the regional consumer price index is

$$P_n = P_n^M,$$

where P_n^M is the Eaton-Kortum price index over goods sourced from all regions.

Assuming that idiosyncratic preference shocks are Type-I extreme value, the choice probabilities differ by age group. For young and middle-aged workers ($a \neq Old$), who have no retirement option, the probability of choosing the market sector in destination region n is

$$\psi_{nm}^{Ma} = \frac{\left(B_n^{Ma} \frac{w_n^{Ma}}{P_n \kappa_{nm}^a} \right)^{1/\eta}}{\sum_{\ell \in N} \left(B_\ell^{Ma} \frac{w_\ell^{Ma}}{P_\ell \kappa_{\ell m}^a} \right)^{1/\eta}}.$$

For old workers ($a = Old$), the retirement sector enters the denominator. The probability of choosing the market sector in destination n is

$$\psi_{nm}^{M,Old} = \frac{\left(B_n^{Ma} \frac{w_n^{Ma}}{P_n \kappa_{nm}^a} \right)^{1/\eta}}{\left(B_m^H \frac{h^a}{P_m} \right)^{1/\eta} + \sum_{\ell \in N} \left(B_\ell^{Ma} \frac{w_\ell^{Ma}}{P_\ell \kappa_{\ell m}^a} \right)^{1/\eta}},$$

and the probability of retiring in the origin region is

$$\psi_m^{H,Old} = \frac{\left(B_m^H \frac{h^a}{P_m} \right)^{1/\eta}}{\left(B_m^H \frac{h^a}{P_m} \right)^{1/\eta} + \sum_{\ell \in N} \left(B_\ell^{Ma} \frac{w_\ell^{Ma}}{P_\ell \kappa_{\ell m}^a} \right)^{1/\eta}}.$$

Market labor of age a employed in destination region n is

$$L_n^{Ma} = \sum_{m \in N} \bar{L}_m^a \psi_{nm}^{Ma}, \quad (1)$$

where ψ_{nm}^{Ma} is given by the age-specific expressions above. Retirement-sector labor in region m is defined only for old workers ($a = Old$):

$$L_m^{Ha} = \bar{L}_m^{Old} \psi_m^{H,Old}.$$

Following [Allen and Arkolakis \(2014\)](#); [Ahlfeldt et al. \(2015\)](#); [Diamond \(2016\)](#), we allow market amenities to respond endogenously to the local mass of market workers:

$$B_n^{Ma} = \bar{B}_n^{Ma} (L_n^{Ma})^{\lambda^a}, \quad B_m^H = \bar{B}_m^H,$$

where $\bar{B}_n^{Ma}$ and $\bar{B}_m^H$ are exogenous amenity components and $\lambda^a \geq 0$ governs the strength of the endogenous market-amenity externality. A positive value of λ^a captures the idea that locations with a larger mass of age- a market workers become more attractive to market workers in the same age group. Retirement-sector amenities are treated as exogenous.

For market sector, we assume that workers within each age group are heterogeneous in their productivity across different tasks. Specifically, a worker ω of age a in region n draws an idiosyncratic productivity shock $z_n^{ia}(\omega)$ for each task i from a Fréchet distribution:

$$F_a(z) = \exp\left(-z^{-\tilde{\theta}_a}\right),$$

where $\tilde{\theta}_a > 1$ is an age-specific shape parameter governing the dispersion of skills across tasks. We allow $\tilde{\theta}_a$ to decline with age (i.e., $\tilde{\theta}_{young} > \tilde{\theta}_{old}$), reflecting the fact that older workers possess more specialized human capital and face higher barriers to switching between task types. A lower $\tilde{\theta}_a$ for the elderly implies a stronger logic of comparative advantage and a more rigid labor supply response to task-specific wage fluctuations.

Each worker supplies labor to the task that maximizes earnings $w_n^{ia} \cdot z_n^{ia}(\omega)$. By the properties of the Fréchet distribution, the efficiency units of task- i labor supplied by age- a workers in region n are

$$\ell_n^{ia} = \Gamma\left(1 - \frac{1}{\tilde{\theta}_a}\right) (w_n^{ia})^{\tilde{\theta}_a - 1} \left[\sum_{j \in I} (w_n^{ja})^{\tilde{\theta}_a} \right]^{\frac{1}{\tilde{\theta}_a} - 1} L_n^{Ma}.$$

Summing earnings across tasks, $w_n^{Ma} L_n^{Ma} = \sum_{i \in I} w_n^{ia} \ell_n^{ia}$, gives the expected market wage:

$$w_n^{Ma} = \Gamma\left(1 - \frac{1}{\tilde{\theta}_a}\right) \left[\sum_{i \in I} (w_n^{ia})^{\tilde{\theta}_a} \right]^{\frac{1}{\tilde{\theta}_a}}. \quad (2)$$

The share of workers in age group a choosing task i in region n is:

$$\tilde{\psi}_n^{ia} = \frac{(w_n^{ia})^{\tilde{\theta}_a}}{\sum_{i' \in I} (w_n^{i'a})^{\tilde{\theta}_a}},$$

and by the law of large numbers the headcount of age- a workers assigned to task i is

$$L_n^{Mia} = \tilde{\psi}_n^{ia} L_n^{Ma}, \quad (3)$$

where L_n^{Ma} is the market labor mass obtained by summing market-sector choices over origins in (1).

3.2 Market Production and Eaton-Kortum Sourcing

In every region, the market sector can produce a continuum of goods indexed by ν . For each good, region m draws an idiosyncratic productivity $z_m(\nu)$ from a Fréchet distribution,

$$F_m(z) = \exp \left[-\mathcal{T}_m z^{-\iota} \right],$$

where $\mathcal{T}_m$ is the Eaton-Kortum technology scale and $\iota > 0$ is the Fréchet shape parameter. Because ι governs how bilateral sourcing shares respond to relative costs, it is also the trade elasticity. This notation replaces the interregional CES elasticity used in the earlier nested-CES sourcing specification.

Conditional on a productivity draw, production combines a task composite and intermediate inputs:

$$q_m(\nu) = z_m(\nu) \left(\frac{\tilde{T}_m}{\beta} \right)^\beta \left(\frac{x_m}{1-\beta} \right)^{1-\beta},$$

where β and $1-\beta$ are the cost shares for the task composite and intermediate inputs, respectively. $\tilde{T}_m$ is the task composite input and x_m is the market final good used as an intermediate input.

The task composite $\tilde{T}_n$ is a CES aggregate over all tasks $i \in I$. We use ϵ to denote the elasticity of substitution between tasks:

$$\tilde{T}_n = \left(\sum_{i \in I} (\tilde{\gamma}^i)^{\frac{1}{\epsilon}} (t_n^i)^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}},$$

where $\tilde{\gamma}^i$ is the demand shifter of task i and t_n^i is the output of task i .

The task composite wage dual to this aggregation is:

$$w_n = \left(\sum_{i \in I} \tilde{\gamma}^i (w_n^i)^{1-\epsilon} \right)^{\frac{1}{1-\epsilon}}.$$

The output of task i is a CES aggregate of effective labor units supplied by all age groups

$a \in G$:

$$t_n^i = \left(\sum_{a \in G} (\tilde{\kappa}^{ia})^{\frac{1}{\tilde{\epsilon}}} (L_n^{Mia})^{\frac{\tilde{\epsilon}-1}{\tilde{\epsilon}}} \right)^{\frac{\tilde{\epsilon}}{\tilde{\epsilon}-1}}.$$

The task efficiency wages are dual to this age-group aggregation:

$$w_n^i = \left(\sum_{a \in G} \tilde{\kappa}^{ia} (w_n^{ia})^{1-\tilde{\epsilon}} \right)^{\frac{1}{1-\tilde{\epsilon}}}.$$

Since there is a single market sector, producers use the market final good as their intermediate input. The intermediate input price index is therefore P_m^M , and the unit cost of the input bundle before the idiosyncratic EK draw is:

$$c_m = (w_m)^\beta (P_m^M)^{1-\beta}.$$

A producer in source region m incurs iceberg trade costs $\tau_{nm} \geq 1$ to ship one unit to destination region n , with $\tau_{nn} = 1$. Under perfect competition, the delivered unit price of good ν from source m to destination n is

$$p_{nm}(\nu) = \frac{\tau_{nm} c_m}{z_m(\nu)}.$$

Destination n purchases each good from the source with the lowest delivered price.

3.3 Endogenous EK Technology and Knowledge Agglomeration

We specify the market-sector EK technology scale as:

$$\mathcal{T}_n = \bar{\mathcal{T}}_n (\Phi_n)^{\gamma_A},$$

where $\bar{\mathcal{T}}_n$ captures the exogenous, location-specific technology component (geography, natural amenities, and historical infrastructure), invariant to demographic shocks. Following the technology diffusion framework of [Coe and Helpman \(1995\)](#); [Eaton and Kortum \(1999\)](#), Φ_n represents the stock of knowledge in region n , which aggregates both localized innovation and cross-regional technology spillovers:

$$\Phi_n = \sum_m \frac{\beta \tilde{\gamma}^{i*} X_{nm} (w_m^{i*}/w_m)^{1-\epsilon}}{w_m^{i*}},$$

where X_{nm} is the bilateral trade flow from source region m to destination region n . Through these trade linkages, n absorbs ideas generated by m 's cognitive workforce.¹

3.4 Final Good Producers

In each destination region, a final good producer purchases each variety ν from the lowest-cost source and aggregates the continuum into a Cobb-Douglas composite:

$$Q_n = \exp\left(\int_0^1 \ln q_n(\nu) d\nu\right).$$

This composite good Q_n serves simultaneously as the household consumption good and as the intermediate input x_n in production, so $x_n = Q_n$. The standard EK result gives the market-sector price index:

$$P_n^M = \Gamma \left[\sum_{m \in \mathcal{N}} \mathcal{T}_m (\tau_{nm} c_m)^{-\iota} \right]^{-\frac{1}{\iota}},$$

where $\mathcal{N}$ includes the 47 Japanese prefectures and the rest of the world, and Γ is a constant that cancels in exact hat algebra. The bilateral expenditure share of destination n on goods sourced from region m is

$$\lambda_{nm} = \frac{\mathcal{T}_m (\tau_{nm} c_m)^{-\iota}}{\sum_{\ell \in \mathcal{N}} \mathcal{T}_\ell (\tau_{n\ell} c_\ell)^{-\iota}}.$$

3.5 Aggregation and General Equilibrium

Goods Market Clearing

Total spending on the market final good in region n is the sum of household final consumption and intermediate input demand:

$$X_n = HE_n + ME_n,$$

where $HE_n = E_n$ is household expenditure on the market good and $ME_n = (1 - \beta)Y_n$ is intermediate input demand. Since there is only one market sector, producers use the market final good as their sole intermediate input, and Cobb-Douglas cost minimization implies that intermediate spending is a constant share of gross sales.

¹In canonical quantitative spatial models (e.g., [Allen and Arkolakis, 2014](#); [Ahlfeldt et al., 2015](#)), endogenous local productivity is typically modeled as a function of total local population or density, standardly expressed as $A_n = \bar{A}_n L_n^{\gamma_A}$. Applying this to a rapidly aging economy is problematic, as a shrinking aggregate population would mechanically imply a nationwide decline in agglomeration externalities. We restrict the endogenous agglomeration force to the NRNM workforce, guided by the empirical evidence in [Section 2](#).

Bilateral trade flows are:

$$X_{nm} = \lambda_{nm} X_n,$$

where the first index denotes the destination and the second index denotes the source.

Goods market clearing requires that total sales from region n equal total spending on n 's goods across all destinations:

$$Y_n = \sum_{m \in \{0,1,\dots,N\}} X_{mn}.$$

Labor Market Clearing

Labor demand comes from variable production for all tasks. Cost minimization gives

$$L_n^{Mia} w_n^{ia} = \beta \tilde{\gamma}^i \left(\frac{w_n^i}{w_n} \right)^{1-\epsilon} Y_n \tilde{\kappa}^{ia} \left(\frac{w_n^{ia}}{w_n^i} \right)^{1-\tilde{\epsilon}}.$$

The labor supply L_n^{Mia} is determined by (3).

In this specification, we assume that while individuals in the retirement sector ($k = H$) consume market goods, their aggregate expenditure does not endogenously respond to the demographic or productivity shocks considered in our analysis. Consequently, the endogenous component of regional expenditure E_n , which drives the market-clearing conditions, is solely comprised of the wage bill from the market-sector workforce:

$$E_n = \sum_{a \in G} \sum_{i \in I} L_n^{Mia} w_n^{ia}.$$

Household spending on the market good is:

$$H E_n = E_n.$$

3.6 General Equilibrium

To quantify the general equilibrium effects of demographic changes, we employ the exact hat algebra method following Dekle et al. (2008). This approach allows us to solve the model in relative changes without needing to calibrate the levels of unobserved fundamentals (e.g., trade costs τ_{nm}). Instead, we condition the analysis on the observed base-year equilibrium outcomes, assuming that these outcomes are rationalized by the underlying fundamentals.

We focus on the comparative static effects of changes in the age composition of the labor force. Formally, for a generic variable x , we define the “hat” notation as $\hat{x} = \frac{x'}{x}$, where x denotes the value in the baseline equilibrium, and x' denotes the value in the counterfactual

equilibrium. For a given set of shocks to resident stocks by origin and age (i.e., $\hat{\bar{L}}_m^a$), the general equilibrium in relative changes is defined as follows.

Definition For given structural parameters, β , λ^a , ι , ϵ , $\tilde{\epsilon}^i$, $\tilde{\theta}_a$, γ_A , and production weights $\tilde{\gamma}^i, \tilde{\kappa}^{ia}$, and for given fundamentals and shocks, $\{\bar{L}_m^a\}$, $\{\tilde{\tau}_m\}$, $\{\tau_{nm}\}$, $\{\mathcal{T}_n\}$, $\{\bar{B}_n^{Ma}\}$, and $\{\bar{B}_m^H\}$, a general equilibrium is given by task efficiency wages $\{\hat{w}_n^{ia}\}$, market price indices $\{\hat{P}_n^M\}$, bilateral sourcing shares $\{\hat{\lambda}_{nm}\}$, total sales $\{\hat{Y}_n\}$, and labor allocations $\{\hat{L}_n^{Mia}\}$, $\{\hat{L}_n^{Ma}\}$, and $\{\hat{L}_m^{Ha}\}$ that satisfy the general equilibrium conditions.

Appendix A shows all of the equilibrium conditions in relative change and the solution algorithm.

3.7 Decomposition of Amenity-Adjusted Market Payoffs

To disentangle the mechanisms driving the general equilibrium effects, specifically how demographic shocks in remote regions transmit to worker payoffs in central hubs, we decompose the change in amenity-adjusted market payoffs for a market sector ($k = M$) worker. Let w_n^{ia} denote the nominal task efficiency wage of a worker of task type i and age group a in region n , let P_n^M denote the market-sector price index, and let B_n^{Ma} denote the endogenous age-specific market amenity. The payoff object is

$$U_n^{Mia} = \frac{w_n^{ia} B_n^{Ma}}{P_n^M}.$$

This object differs from a real wage because it includes the location-choice amenity generated by the local market labor mass of age group a . Agricultural prices respond to land rents and rural labor supply, channels that are distinct from the agglomeration and trade mechanisms of interest here. We therefore deflate by the market-sector price index and add the endogenous amenity channel explicitly.

The change in amenity-adjusted market payoffs can be decomposed as follows:

$$\begin{aligned} \ln \hat{U}_n^{Mia} = & \underbrace{\ln \hat{w}_n^{ia}}_{\text{Own Scarcity}} + \underbrace{\ln \hat{B}_n^{Ma}}_{\text{Amenity}} \\ & + \frac{1}{\iota} \sum_m \lambda_{nm} \underbrace{\gamma_A \ln \hat{\Phi}_m}_{\text{Technical Access}} \\ & - \sum_m \lambda_{nm} \beta \underbrace{\left[\sum_i \Omega_m^i \ln \hat{w}_m^i \right]}_{\text{Task Cost}} \\ & - \sum_m \lambda_{nm} \underbrace{\left[(1 - \beta) \ln \hat{P}_m^M + \ln \hat{\tau}_{nm} \right]}_{\text{Non-Labor Cost}}. \end{aligned}$$

where Ω_m^i is the cost share of task i in the market sector.

The own-scarcity term represents the direct labor market effect of demographic shifts on own wages. In scenarios where young labor becomes relatively scarce, this term is positive, reflecting the standard Rybczynski effect as the marginal product of the scarce factor rises.

The amenity term captures the location-choice channel. When a region retains more workers of age a , the endogenous age-specific market amenity rises relative to regions where that group contracts more sharply. This term is central for interpreting migration responses, since workers choose locations based on amenity-adjusted market payoffs rather than wages alone.

The technical-access term is endogenous knowledge diffusion, $\gamma_A \ln \hat{\Phi}_m$, in source region m . Because $\mathcal{T}_m$ is the technology scale, its contribution to the log price index is scaled by $1/\iota$. Through trade linkages (λ_{nm}), region n benefits from productivity gains in all regions it sources from. When m has stronger access to the NRNM knowledge stock, region n captures a share of that gain proportional to λ_{nm} .

The task-cost term represents the task cost effect within the supply component. It decomposes unit labor costs into a weighted average of task-specific wages. Under gross complementarity ($\epsilon < 1$), an increase in the wage of a scarce factor raises average production costs more than proportionally. Through spatial input-output linkages, central regions are exposed to wage increases in aging peripheral regions. If peripheral areas supply intermediate goods produced by relatively expensive routine labor, the resulting rise in intermediate prices reduces amenity-adjusted payoffs in central hubs.

The non-labor cost term captures changes in intermediate input costs (the market-sector price index $\hat{P}_m^M$ weighted by the intermediate input share $1 - \beta$) and bilateral trade frictions $\hat{\tau}_{nm}$. Retirement-sector price dynamics are excluded from this decomposition by construction, since the retirement sector generates no market output and carries no price index in this framework, and market payoffs are deflated by the market-sector price index P_n^M .

Appendix C shows the derivation of the payoff decomposition.

4 Quantification

We model the Japanese economy as a small open economy relative to the RoW. Given that demographic changes within Japan are unlikely to exert significant general equilibrium effects on the global wage structure, and due to the lack of granular age-task-sector data for the global aggregate, we treat the foreign labor market as exogenous. Specifically, we choose the RoW wage as the numeraire, normalizing the change in foreign wages to unity ($\hat{w}_{RoW} = 1$). The foreign price index is still determined inside the EK system because foreign producers use the market final good as an intermediate input. Domestic supply shocks in Japan therefore transmit to foreign production costs through input-output linkages, even though foreign wages

are fixed.

We bring the model to the data from the Japanese economy and the RoW. Our analysis covers 47 prefectures of Japan. All input-output shares, labor cost shares, and intermediate input shares are aggregated accordingly from the Japanese Inter-Prefectural Input-Output Tables and the WIOD.

Our analysis relies on a combination of national, labor-market, and migration datasets to construct and calibrate the spatial general equilibrium model.

Input-Output and Trade Data: We use the World Input-Output Database and the Japanese Inter-Prefectural Input-Output Tables to capture inter-industry relationships, global supply chain dynamics, and regional trade flows. These tables provide the fundamental data for sectoral production, intermediate input shares, and final demand across 47 prefectures of Japan and the RoW.

Labor Market and Task Data: To map workers to tasks and sectors, we utilize the Japanese Employment Status Survey and the Population Census. These datasets provide the distribution of employment by age, sector, and occupation. Crucially, to characterize the task content of occupations, we link these data with task descriptors from the ONET database. Following the literature (Doi and Suzuki 2025a; Acemoglu and Autor 2011), we classify occupations into four broad task categories: Task 1 (Non-Routine Non-Manual), Task 2 (Non-Routine Manual), Task 3 (Routine Non-Manual), and Task 4 (Routine Manual). This mapping allows us to construct the age-task-sector employment matrices required by our model. Wage data by age and occupation are sourced from the Survey on Wage Structure.

Migration Data: We construct baseline interprefectural migration shares from the 2020 Population Census (*Population Census of Japan*), published by the Statistics Bureau of Japan. The census records the prefecture of residence five years prior, allowing us to construct origin-destination migration flows for the 2015–2020 period. We aggregate these flows across the 47 prefectures into the three model age groups and normalize flows within each origin-age cell to obtain the migration shares used in the location-choice block.

4.1 Calibration

Below, we outline the calibration of structural parameters.

The labor cost share β and the intermediate input share $1 - \beta$ are directly calibrated from the input-output tables (IPIO and WIOD) by computing the relevant cost shares in the baseline year.

We set the EK trade elasticity to $\iota = 4$. In the model, ι is the Frechet shape parameter of regional productivity draws and governs the elasticity of bilateral sourcing shares with respect to delivered costs. This value lies in the range commonly used in quantitative trade models and is close to estimates surveyed by Hillberry and Hummels (2013) and used in Eaton-Kortum applications such as Caliendo and Parro (2015).

We set the migration choice elasticity $1/\eta = 0.6$ based on estimates from recent literature on internal migration in Japan, such as [Doi and Suzuki \(2025b\)](#) and [Suzuki \(2023\)](#), typically ranging between 0.5 and 2.

The parameter $\tilde{\theta}_a$ governs the heterogeneity of worker productivity across tasks and thus the elasticity of task supply with respect to wage differentials. We allow it to vary across age groups, reflecting the well-documented decline in occupational mobility with age. For young workers ($a = \text{young}$) and middle-aged workers ($a = \text{middle}$), we follow [Burstein et al. \(2019\)](#) and set $\tilde{\theta}_{\text{young}} = \tilde{\theta}_{\text{middle}} = 1.8$, calibrated to match the elasticity of occupational sorting with respect to wage changes. For elderly workers ($a = \text{old}$), we set $\tilde{\theta}_{\text{old}} = 1.1$. This lower value reflects the accumulation of task-specific human capital over the career: older workers have a stronger comparative advantage in their current task and face larger implicit costs of switching to a different task type. [Gathmann and Schönberg \(2010\)](#) document that inter-task mobility among workers aged 55–64 is approximately 60 percent that of workers aged 25–34, implying a ratio $\tilde{\theta}_{\text{old}}/\tilde{\theta}_{\text{young}} \approx 0.6$; with $\tilde{\theta}_{\text{young}} = 1.8$ this yields $\tilde{\theta}_{\text{old}} \approx 1.1$. The lower bound $\tilde{\theta}_a > 1$ is required for the Fréchet distribution to have a finite mean. [Dix-Carneiro \(2014\)](#) provides further support, showing that sector-switching costs rise steeply with age, consistent with a reduced task reallocation elasticity for elderly cohorts.

We set the elasticity of substitution between different age groups within a task to $\tilde{\epsilon}^i = 0.5$. While early macroeconomic aggregate studies (e.g., [Card and Lemieux \(2001\)](#)) often estimate the elasticity of substitution across age cohorts to be greater than one, these estimates implicitly assume workers of different ages perform identical tasks with varying efficiencies. In our task-based spatial framework, age cohorts specialize in fundamentally different tasks (e.g., fluid-intelligence-intensive tasks for the young versus experience-based cognitive tasks for the old). Furthermore, recent firm-level evidence confirms that workers of different age groups act as strong complementary factors in production. As empirically demonstrated by [Wasmer \(2001\)](#), the elasticity of substitution between age cohorts with different knowledge and experience levels is approximately 0.5.

We set the elasticity of substitution between tasks to $\epsilon = 0.9$. This calibration is directly informed by the structural estimates of [Goos et al. \(2014\)](#) in their analysis of labor market polarization. A value below unity is critical as it implies that routine and non-routine tasks are gross complements, consistent with the theoretical task-based frameworks of [Acemoglu and Restrepo \(2019\)](#) and [Lee and Shin \(2017\)](#).

Following the structural framework of [Ahlfeldt et al. \(2015\)](#), we calibrate the productivity spillover elasticity to $\gamma_A = 0.07$. This value represents a balanced middle ground in the quantitative spatial literature. While [Moretti \(2004\)](#) identifies stronger human capital externalities near 0.50 for cognitive tasks, we adopt a more conservative γ_A to ensure global stability and prevent unrealistic concentration in major hubs. This choice also remains more responsive to local density than the 0.02 used by [Suzuki \(2023\)](#), reflecting the high knowledge intensity of

NRNM activities.

The elasticity of local age-specific market amenities with respect to the regional market population size of cohort a , λ^a , is set to 0.15 in the baseline quantitative exercise. This parameterization is grounded in the empirical urban economics literature on local consumption amenities and demographic-specific private goods. Schiff (2015) shows that local consumer variety rises concavely with city population, while Waldfogel (2008) documents that the supply of private amenities responds to the absolute size of the relevant consumer group.

5 Counterfactual Analysis

The central question motivating our counterfactual is: what would the Japanese economy look like if the demographic structure in each prefecture, as observed in 2015, were to shift to the distribution projected for 2045? To quantify the aggregate and spatial implications of this transition, we ground our counterfactual exercises in the official demographic projections provided by the National Institute of Population and Social Security Research. Taking the year 2015 as our demographic baseline, we simulate the projected shift in the workforce age structure for 2045. We construct the shocks from the IPSS all-prefectures-and-municipalities file, *Estimated Future Population by Sex and Age Group (5 Years) by Prefecture and Municipality Assuming Blocked Population*. This series assumes no migration and therefore captures natural population change within each prefecture, driven by the baseline age structure and projected survival and fertility patterns.

We implement the demographic transition as an age-by-prefecture labor supply shock. Let $\hat{L}_n^a$ denote the ratio of projected 2045 population to baseline 2020 population for age group a in prefecture n . The model uses the full 3×47 matrix of regional shocks, allowing population aging to differ across space rather than imposing a uniform national age-composition shift. To keep the main text compact, Table 2 reports the average shock across prefectures for each age group, together with the regional range.

Table 2: Projected Regional Demographic Shocks for 2045 under the Blocked-Population Assumption

Age Cohort	2015 Working Population	Mean Shock	Minimum	Maximum
Young (15–34)	18,540,464	0.769	0.590	1.025
Middle (35–64)	39,199,676	0.711	0.579	0.846
Senior (65+)	30,751,548	1.076	0.855	1.543

Note: Population data and regional projections are from the National Institute of Population and Social Security Research *Estimated Future Population by Sex and Age Group (5 Years) by Prefecture and Municipality Assuming Blocked Population*. The shock is defined as $\hat{L}_n^a = L_{n,2045}^a / L_{n,2015}^a$. Mean, minimum, and maximum are computed across the 47 prefectures. The counterfactual simulations use the full age-by-prefecture shock matrix.

We then introduce an automation extension through task demand shifters. In the model, $\hat{\gamma}^i$ governs the contribution of task i to market sector output. The automation extension reduces the demand shifters for routine tasks by 10 percent:

$$\hat{\gamma}^{RNM} = \hat{\gamma}^{RM} = 0.90, \quad \hat{\gamma}^{NRNM} = \hat{\gamma}^{NRM} = 1.$$

This specification treats automation as a routine-task displacement shock rather than as a direct increase in NRNM demand. It is consistent with the task displacement framework of Autor et al. (2003); Acemoglu and Restrepo (2019). The automation shock differs from both the demographic shock and the productivity shifter: the demographic shock changes the age composition of labor supply, the productivity shifter changes regional TFP through task exposure, and the automation shock changes firms' relative demand for routine task inputs.

We therefore run two scenarios:

1. **Aging**: the 2045 demographic shock from Table 2.
2. **Aging + automation**: the same demographic shock, plus a 10 percent decline in the demand shifters for RNM and RM tasks.

5.1 Results: Aging Shock

Output Growth Channels

Figure 6 plots the simulated changes in regional output against the baseline share of older workers under endogenous productivity. Panel (a) colors prefectures by their dominant baseline task specialization. Panels (b) and (c) use the same vertical outcome but color each prefecture by the productivity change and the young market-labor change, respectively. Large squares denote metropolitan hubs.

The output ranking is not determined by old-age exposure alone. Tokyo, Okinawa, Fukuoka, and Osaka remain at the top of the output distribution, with output changes of 1.215, 1.253, 1.125, and 1.083. These regions also have the strongest young market-labor retention. Tokyo's young market labor falls by only 4.7 percent, compared with much larger declines in shrinking prefectures such as Fukushima and Akita. The same regions receive larger productivity support through the NRNM absorption channel, as shown in Panel (b).

This production ranking does not imply the same ranking for worker payoffs. Worker incidence depends on age-task scarcity, endogenous amenities, and cost pass-through rather than output alone. The next subsection makes this distinction explicit by comparing Tokyo, the main production and retention center, with Akita, the lowest-output prefecture in the counterfactual.

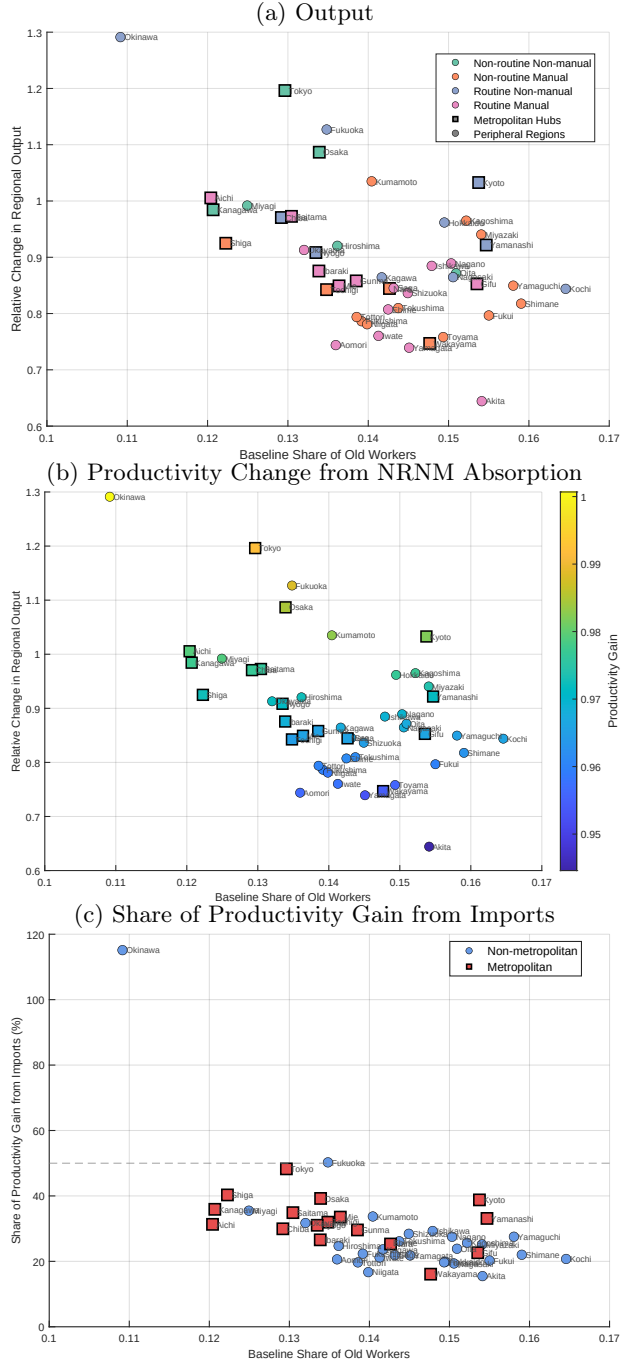


Figure 6: Regional Output Growth Channels

Note: Colors indicate each prefecture's dominant baseline task specialization; squares denote metropolitan hubs. Metropolitan prefectures are defined following the Statistics Bureau of Japan's 2020 Population Census metropolitan area classification. The three major metropolitan areas comprise the Greater Tokyo Area (Tokyo, Kanagawa, Saitama, Chiba, Ibaraki, Tochigi, Gunma, and Yamanashi), the Greater Nagoya Area (Aichi, Gifu, and Mie), and the Greater Osaka Area (Osaka, Kyoto, Hyogo, Nara, Shiga, and Wakayama). All remaining prefectures are classified as non-metropolitan.

Amenity-Adjusted Real Wages

Table 3 decomposes amenity-adjusted real wage changes for Tokyo and Akita by age group and task. The first two terms vary by worker type and age: local scarcity is task-age specific, while the amenity term is age-region specific. The remaining three terms are destination-level general-equilibrium price-index components and are therefore common to all age-task groups within a prefecture.

Table 3: Detailed Decomposition of Amenity-Adjusted Real Wage Changes by Mechanism

Region	Age Group	Task Type	Term 1 (Local Scarcity)	Term 2 (Amenity)	Term 3 (Tech. Access)	Term 4 (Task Cost)	Term 5 (Non-Labor)	Total (Amenity-Adjusted Real Wage)
Tokyo	Young (15-34)	Non-routine Non-manual	-0.0903					-0.2086
	Young (15-34)	Non-routine Manual	-0.0871	-0.0036				-0.2053
	Young (15-34)	Routine Non-manual	-0.0870					-0.2052
	Young (15-34)	Routine Manual	-0.0858					-0.2040
	Middle (35-64)	Non-routine Non-manual	0.0140					-0.1121
	Middle (35-64)	Non-routine Manual	0.0173	-0.0115	-0.0027	0.0057	-0.1177	-0.1088
	Middle (35-64)	Routine Non-manual	0.0174					-0.1087
	Middle (35-64)	Routine Manual	0.0186					-0.1076
	Senior (65+)	Non-routine Non-manual	-0.5681					-0.6828
	Senior (65+)	Non-routine Manual	-0.5635	0.0000				-0.6781
	Senior (65+)	Routine Non-manual	-0.5633					-0.6780
	Senior (65+)	Routine Manual	-0.5616					-0.6763
Akita	Young (15-34)	Non-routine Non-manual	0.1217					-0.1736
	Young (15-34)	Non-routine Manual	0.1233	-0.1189				-0.1719
	Young (15-34)	Routine Non-manual	0.1258					-0.1695
	Young (15-34)	Routine Manual	0.1253					-0.1700
	Middle (35-64)	Non-routine Non-manual	0.0796					-0.2125
	Middle (35-64)	Non-routine Manual	0.0812	-0.1157	-0.0110	-0.0221	-0.1432	-0.2109
	Middle (35-64)	Routine Non-manual	0.0836					-0.2084
	Middle (35-64)	Routine Manual	0.0831					-0.2090
	Senior (65+)	Non-routine Non-manual	-0.5459					-0.7223
	Senior (65+)	Non-routine Manual	-0.5436	0.0000				-0.7199
	Senior (65+)	Routine Non-manual	-0.5400					-0.7164
	Senior (65+)	Routine Manual	-0.5408					-0.7172

Note: Term 1 isolates the direct local scarcity effect on nominal task wages. Term 2 is the endogenous age-specific market amenity, computed as $\ln \hat{B}_n^{Ma} = \lambda^a \ln \hat{L}_n^{Ma}$ with $\lambda^a = 0.15$ for young and middle-aged workers and $\lambda^a = 0$ for senior workers. Terms 3 through 5 are the general-equilibrium price-index components shared within destination regions: Term 3 captures technical access, Term 4 reflects spatial transmission of task labor costs, and Term 5 combines intermediate-input costs and trade frictions. The total is the log change in amenity-adjusted real wage, $\ln \hat{U}_n^{Mia} = \ln \hat{w}_n^{ia} + \ln \hat{B}_n^{Ma} - \ln \hat{P}_n^M$.

Table 3 shows that aging produces welfare losses for all age groups in both regions. For young workers, the local scarcity component is small and negative in Tokyo (approximately -0.005) and moderately positive in Akita (approximately 0.163). In Tokyo, the non-labor cost term subtracts about 0.122 log points and the amenity loss adds a further 0.007 , yielding total welfare changes of roughly -0.126 to -0.124 log points. In Akita, the positive scarcity term is more than offset by an amenity loss of 0.122 log points and the remaining cost channels, leaving young workers with changes of roughly -0.136 to -0.133 log points. Both regions therefore see young workers worse off, with Akita registering slightly larger losses despite the higher nominal scarcity component.

The pattern holds for middle-aged workers, with wider regional dispersion. In Tokyo, the scarcity term for middle-aged workers is negative (approximately -0.038), so neither scarcity

nor amenity provides any offset, and the total falls to around -0.156 to -0.154 log points. In Akita, the middle-aged scarcity component is positive (approximately 0.041) but smaller than the amenity loss of 0.112 log points, and the general-equilibrium cost channels add further drag, pushing the total to roughly -0.248 to -0.246 log points. The gap between regions is wider at the middle-age margin than at the young margin, driven by Akita’s larger amenity elasticity response.

Senior workers face the largest losses. Their local scarcity components are sharply negative, approximately -0.304 in Tokyo and -0.298 in Akita, because senior labor supply expands relative to demand. Because the amenity elasticity for senior workers is zero ($\lambda^a = 0$), the amenity term is absent, and losses reflect only the scarcity and general-equilibrium cost channels. Total welfare changes are approximately -0.418 to -0.415 in Tokyo and -0.475 to -0.471 in Akita. The regional gap for seniors reflects differences in cost pass-through rather than amenities. Variation across tasks within an age group remains small relative to variation across age groups and regions, so the main incidence margins are age and place rather than task type within age.

5.2 Results: Aging + Automation Shock

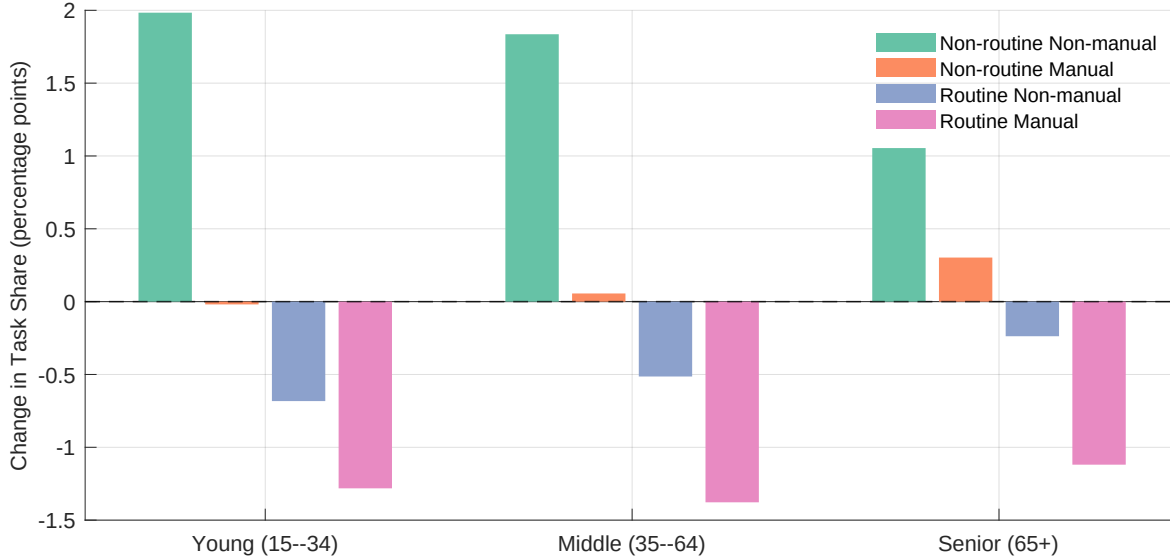


Figure 7: Task Reallocation

Note: Each panel reports the incremental effect of the automation extension relative to the aging + technology counterfactual.

Figure 7 shows how the automation extension reshapes labor allocation across age groups. Because the automation shock lowers the demand shifters for routine tasks, it works mainly through reallocation rather than through an additional productivity gain. Panel (a) shows whether workers remain in market production but move away from RNM and RM tasks. Panel

(b) shows whether the routine-task contraction is large enough to reduce market participation.

The reallocation pattern is clear. For all three age groups, the largest decline falls on routine non-manual work, while routine manual work also contracts. The displaced labor is absorbed by the two non-routine categories. Young workers shift 1.03 percentage points out of RNM tasks and 0.58 percentage points out of RM tasks, with corresponding gains of 1.08 percentage points in NRNM and 0.53 percentage points in NRM work. Middle-aged workers gain 1.06 percentage points in NRNM work and 0.46 percentage points in NRM work, while losing 0.70 percentage points in RNM and 0.82 percentage points in RM. Senior workers move in the same direction, gaining 0.98 percentage points in NRNM and 0.49 percentage points in NRM while leaving routine tasks.

Panel (b) shows that automation does not simply reshuffle employment within the market sector. It also reduces market participation for every age group. The decline is largest for young workers, at 2.52 percentage points, followed by 1.68 percentage points for middle-aged workers and 1.21 percentage points for senior workers. In this calibration, the automation extension therefore operates through two margins at once. It reallocates labor away from routine tasks, and it pushes some workers out of market work entirely. The first effect is common across age groups. The second is strongest for the young.

6 Conclusion

Population aging in Japan is not a shock that falls uniformly across space. Its spatial footprint is determined by the task content of production. Because age cohorts differ systematically in their task capabilities, a shift in the age composition of the workforce changes the effective supply of each task type, with consequences that vary across sectors and regions according to how intensively each task is used locally.

The counterfactual analysis yields three findings. First, the scarcity of young cognitive workers interacts with endogenous technical access. Regions with high baseline exposure to NRNM-intensive production, especially Tokyo, convert the demographic transition into stronger output performance and retain more young market labor. Second, production incidence and worker incidence differ. Young labor retention supports output in metropolitan areas, but it also compresses the local scarcity premium for young NRNM workers. The largest percentage payoff gains for young cognitive workers appear in some non-metropolitan regions where young labor becomes much scarcer. Third, senior workers remain the most exposed group under the aging and technology counterfactual. Senior payoffs fall nearly everywhere because the own-wage component is weak and cost channels reduce purchasing power. The welfare consequences of aging are therefore distributed asymmetrically across both space and generations.

The automation extension adds an additional layer of adjustment. By lowering demand for routine tasks, it induces reallocation toward non-routine work across all age groups and

modestly reduces market participation, with the largest exit response among the young. In our framework, automation therefore amplifies the task-composition consequences of aging even when its direct effect operates through task demand rather than through an additional productivity shock.

These findings point to two policy implications. Measures that expand the supply of young cognitive workers in metropolitan areas, whether through education, migration, or spatial policy, can moderate the productivity gap between cores and periphery. Measures that create task-appropriate employment for older workers in both urban and peripheral settings can limit the amenity-adjusted payoff losses that older cohorts otherwise bear. The results also suggest that aggregate indicators of labor supply adequacy that focus on headcounts may understate structural mismatches arising from the task composition of the workforce, and that regional policy in aging societies should account for the task dimension of local comparative advantage. Incorporating endogenous technology adoption and immigration into this framework are natural extensions for future work.

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Appendix

A Solution Algorithm

We solve the model in relative changes using the following iterative procedure.

1. Guess a vector of relative changes for task efficiency wages $\{\hat{w}_n^{ia}\}_{n,i,a}$, total market sales $\{\hat{Y}_n\}_n$, market price indices $\{\hat{P}_n^M\}_n$, bilateral trade flows $\{\hat{X}_{nm}\}_{n,m}$, and labor distribution $\{\hat{L}_n^{ia}\}_{n,i,a}$.

A. Solve Prices and Unit Costs:

- Compute task wage index $\hat{w}_n^i$ from age-group wages using (F0).
- Compute market sector wage index $\hat{w}_n$ using (F1).
- Compute technical-access exposure, accessible NRNM knowledge stock $\hat{\Phi}_n$, and EK technology scale $\hat{\mathcal{T}}_n$ using (F10).
- Compute unit cost $\hat{c}_n$ using (F2).
- Compute EK sourcing term $\hat{\mathcal{D}}_{nm}$ using (F3).
- Update market sector price index $\hat{P}_n^M$ using (F4).
- Compute consumer price index $\hat{P}_n$ using (H4).
- Iterate Step A until price indices converge.

B. Solve Sales and Wages:

- Compute total regional expenditure $\hat{E}_n = \hat{E}_n^M$ using (H8) and (H9).
- Compute bilateral trade share changes $\hat{\lambda}_{nm}$ using (F5).
- Compute intermediate demand $\hat{M}E_n$ using (F6), based on the guessed $\hat{Y}_n$.
- Compute total market spending $\hat{X}_n$ using (F7).
- Compute bilateral trade flows $\hat{X}_{nm}$ using (F8).
- Update total sales $\hat{Y}_{n,\text{new}}$ using (F9).
- Compute model-implied labor demand from (H1). Update task efficiency wages $\hat{w}_{n,\text{new}}^{ia}$ via Walrasian adjustment until labor markets clear.
- Normalize the change in foreign wages to unity: $\hat{w}_{\text{RoW}}^{ia} = 1$.
- Iterate Step B until sales and market wages converge.

C. Solve Labor Supply:

- Compute age-group market wage index $\hat{w}_n^{Ma}$ using (H2).
- Compute market amenity changes $\hat{B}_n^{Ma}$ and set retirement-sector amenity changes $\hat{B}_n^{Ha} = 1$ using (H0).
- Compute origin-specific market location-sector choices $\hat{\psi}_{nm}^{Ma}$ and retirement-sector choices $\hat{\psi}_m^{Ha}$ using (H3).

- Update market-sector labor supply $\hat{L}_n^{Ma}$ and retirement-sector labor supply $\hat{L}_m^{Ha}$ using (H5).
 - Compute task selection shares $\hat{\psi}_n^{ia}$ using (H6).
 - Update task-specific labor supply $\hat{L}_n^{Mia}$ using (H7).
2. Check for convergence. If $\{\hat{L}_{n,\text{new}}^{ia}\} \approx \{\hat{L}_n^{ia}\}$ and $\{\hat{X}_{nm,\text{new}}\} \approx \{\hat{X}_{nm}\}$, stop. Otherwise, update all guesses with a dampening factor and return to Step 1.

B Alternative Production Structure: Capital-Task Complementarity

B.1 Framework

In the main text, we adopted a standard production function to focus on the general equilibrium effects of demographic changes. However, a growing body of literature suggests that capital and automation technologies do not substitute for all labor types equally. They tend to substitute for routine tasks while complementing non-routine cognitive tasks (Autor et al. 2013). Since age groups in Japan differ systematically in their task exposure, a canonical Cobb-Douglas specification may understate the heterogeneous impacts of automation.

To address this, we present an alternative supply-side framework for the single market sector. We employ a nested CES production function (Borjas 2003; Ottaviano and Peri 2012) augmented with a task-based structure (Acemoglu and Autor 2011). Capital (k_n) enters a specific nest where it complements non-routine cognitive labor and substitutes for routine labor. This formalizes the mechanism by which a decline in the price of capital raises relative demand for non-routine cognitive labor while reducing demand for routine labor.

The production function $q_n(\omega)$ has four aggregation layers. The task input t_n^i aggregates age-group labor with elasticity $\tilde{\epsilon}^i$:

$$t_n^i = \left(\sum_{a \in G} (\tilde{\kappa}^{ia})^{\frac{1}{\tilde{\epsilon}}} (L_n^{ia})^{\frac{\tilde{\epsilon}-1}{\tilde{\epsilon}}} \right)^{\frac{\tilde{\epsilon}}{\tilde{\epsilon}-1}},$$

where L_n^{ia} is effective labor of age group a in task i .

Tasks are partitioned into the complementary composite T_n^1 (non-routine cognitive, I_1) and the substitutable composite T_n^2 (routine tasks, I_2):

$$T_n^1 = \left(\sum_{i \in I_1} (\tilde{\gamma}^i)^{\frac{1}{\epsilon^1}} (t_n^i)^{\frac{\epsilon^1-1}{\epsilon^1}} \right)^{\frac{\epsilon^1}{\epsilon^1-1}}, \quad T_n^2 = \left(\sum_{i \in I_2} (\tilde{\gamma}^i)^{\frac{1}{\epsilon^2}} (t_n^i)^{\frac{\epsilon^2-1}{\epsilon^2}} \right)^{\frac{\epsilon^2}{\epsilon^2-1}}.$$

The value-added composite uses a nested structure (Parro 2013; Krusell et al. 2000). In

Table A1: Equilibrium Conditions

(H0)	$\hat{B}_n^{Ma} = \left(\hat{L}_n^{Ma}\right)^{\lambda^a}, \quad \hat{B}_n^{Ha} = 1$	$\forall(n, a)$
(H1)	$\hat{L}_n^{ia} \hat{w}_n^{ia} = \hat{Y}_n \hat{\gamma}^i \left(\frac{\hat{w}_n^i}{\hat{w}_n}\right)^{1-\epsilon} \left(\frac{\hat{w}_n^{ia}}{\hat{w}_n^i}\right)^{1-\bar{\epsilon}^i}$	$\forall(n, i, a)$
(H2)	$\hat{w}_n^{Ma} = \left[\sum_i \tilde{\psi}_n^{ia} (\hat{w}_n^{ia})^{\bar{\theta}_a}\right]^{\frac{1}{\bar{\theta}_a}}$	$\forall(n, a)$
(H3)	$\mathcal{S}_m^a = \psi_m^{Ha} \left(\frac{\hat{B}_m^{Ha} \hat{h}_m^a}{\hat{P}_m}\right)^{1/\eta} + \sum_\ell \psi_{\ell m}^{Ma} \left(\frac{\hat{B}_\ell^{Ma} \hat{w}_\ell^{Ma}}{\hat{P}_\ell \hat{\kappa}_{\ell m}^a}\right)^{1/\eta},$	$\forall(n, m, a)$
(H4)	$\hat{\psi}_{nm}^{Ma} = \frac{\left(\frac{\hat{B}_n^{Ma} \hat{w}_n^{Ma}}{\hat{P}_n \hat{\kappa}_{nm}^a}\right)^{1/\eta}}{\mathcal{S}_m^a}, \quad \hat{\psi}_m^{Ha} = \frac{\left(\frac{\hat{B}_m^{Ha} \hat{h}_m^a}{\hat{P}_m}\right)^{1/\eta}}{\mathcal{S}_m^a}$	$\forall(n, m, a)$
(H5)	$\hat{P}_n = \hat{P}_n^M$	$\forall(n)$
(H6)	$\hat{L}_n^{Ma} = \sum_m \xi_{nm}^{Ma} \hat{L}_m^a \hat{\psi}_{nm}^{Ma},$	$\forall(n, a); \forall(m, a)$
(H7)	$\hat{L}_m^{Ha} = \hat{L}_m^a \hat{\psi}_m^{Ha}$	$\forall(m, a)$
(H8)	$\hat{\psi}_n^{ia} = \frac{(\hat{w}_n^{ia})^{\bar{\theta}_a}}{\sum_j \tilde{\psi}_n^{ja} (\hat{w}_n^{ja})^{\bar{\theta}_a}} = (\hat{w}_n^{ia} / \hat{w}_n^{Ma})^{\bar{\theta}_a}$	$\forall(n, i, a)$
(H9)	$\hat{L}_n^{Mia} = \hat{\psi}_n^{ia} \hat{L}_n^{Ma}$	$\forall(n, i, a)$
(H10)	$\hat{E}_n = \sum_{i,a} \chi_n^{Mia} \hat{L}_n^{Mia} \hat{w}_n^{ia}$	$\forall(n)$
(F1)	$\hat{E}_n^M = \hat{E}_n$	$\forall(n)$
(F2)	$\hat{w}_n^i = \left(\sum_a \Psi_n^{ia} (\hat{w}_n^{ia})^{1-\bar{\epsilon}}\right)^{\frac{1}{1-\bar{\epsilon}}}$	$\forall(n, i)$
(F3)	$\hat{w}_n = \left(\sum_i \theta_n^i \hat{\gamma}^i (\hat{w}_n^i)^{1-\epsilon}\right)^{\frac{1}{1-\epsilon}}$	$\forall(n)$
(F4)	$\hat{c}_n = (\hat{w}_n)^\beta \left(\hat{P}_n^M\right)^{1-\beta}$	$\forall(n)$
(F5)	$\hat{\mathcal{D}}_{nm} = \hat{\mathcal{T}}_m (\hat{\tau}_{nm} \hat{c}_m)^{-\iota}$	$\forall(n, m)$
(F6)	$\hat{P}_n^M = \left[\sum_m \lambda_{nm} \hat{\mathcal{D}}_{nm}\right]^{-\frac{1}{\iota}}$	$\forall(n)$
(F7)	$\hat{\lambda}_{nm} = \frac{\hat{\mathcal{D}}_{nm}}{\sum_\ell \lambda_{n\ell} \hat{\mathcal{D}}_{n\ell}}$	$\forall(n, m)$
(F8)	$\hat{M}E_n = \hat{Y}_n$	$\forall(n)$
(F9)	$\hat{X}_n = \frac{E_n}{X_n} \hat{E}_n + \frac{ME_n}{X_n} \hat{M}E_n$	$\forall(n)$
(F10)	$\hat{X}_{nm} = \hat{\lambda}_{nm} \hat{X}_n$	$\forall(n, m)$
(F11)	$\hat{Y}_n = \sum_m \frac{X_{mn}}{Y_n} \hat{X}_{mn}$	$\forall(n)$
(F12)	$\hat{\mathcal{T}}_n = \left(\hat{\Phi}_n\right)^{\gamma^A}, \quad \hat{\Phi}_n = \sum_m \phi_{nm}^{i*} \left[\hat{X}_{nm} (\hat{w}_m^{i*})^{-\epsilon} (\hat{w}_m)^{\epsilon-1}\right]$	$\forall(n)$

Table A2: Baseline Fundamentals

Symbol	Description	Dimension	Construction / Definition
λ_{nm}	Bilateral trade share	$N \times N$	$\frac{X_{nm}}{\sum_{m'} X_{nm'}}$
$\frac{E_n}{X_n}$	Final consumption ratio	N	$\frac{\text{Final Demand}_n}{\text{Total Output}_n}$
$\frac{ME_n}{X_n}$	Intermediate use ratio	N	$1 - \frac{E_n}{X_n}$
$\frac{X_{nm}}{Y_m}$	Sales destination share	$N \times N$	$\frac{X_{nm}}{\sum_n X_{nm}}$
θ_n^i	Task i cost share in production	$T \times N$	$\frac{\tilde{\gamma}^i (w_n^i/w_n)^{1-\epsilon}}{\sum_{i'} \tilde{\gamma}^{i'} (w_n^{i'}/w_n)^{1-\epsilon}}$
Ψ_n^{ia}	Age a share of task i wage bill	$A \times N \times T$	$\frac{L_n^{Mia} w_n^{ia}}{L_n^{Mi} w_n^i}$
$\tilde{\psi}_n^{ia}$	Task i share within market age- a labor	$A \times N \times T$	$\frac{L_n^{Mia}}{L_n^{Ma}}$
χ_n^{Mia}	Market income share of task i , age a	$A \times N \times T$	$\frac{L_n^{Mia} w_n^{ia}}{E_n}$
$\bar{L}_m^a$	Initial resident stock by origin and age	$A \times N$	Baseline residents of age group a initially located in origin m
ψ_{nm}^{Ma}	Initial market choice share for destination n and origin m	$A \times N \times N$	$\frac{L_{nm}^{Ma}}{\bar{L}_m^a}$
ψ_m^{Ha}	Initial retirement-sector choice share for origin m ($a = Old$ only)	$A \times N$	$\frac{L_m^{Ha}}{\bar{L}_m^a}$
ξ_{nm}^{Ma}	Origin contribution share to destination market labor	$A \times N \times N$	$\frac{\bar{L}_m^a \psi_{nm}^{Ma}}{L_n^{Ma}}$
ϕ_{nm}^{i*}	Accessibility weight for knowledge stock	$N \times N$	$\frac{X_{nm} (w_m^{i*}/w_m)^{1-\epsilon} / w_m^{i*}}{\sum_{m'} X_{nm'} (w_{m'}^{i*}/w_{m'})^{1-\epsilon} / w_{m'}^{i*}}$

the inner nest (h_n), capital combines with T_n^1 with a low elasticity $\sigma_1 < 1$:

$$h_n = \left[\delta_n^{\frac{1}{\sigma_1}} k_n^{\frac{\sigma_1-1}{\sigma_1}} + (1 - \delta_n)^{\frac{1}{\sigma_1}} (T_n^1)^{\frac{\sigma_1-1}{\sigma_1}} \right]^{\frac{\sigma_1}{\sigma_1-1}}.$$

In the outer nest (v_n), the bundle h_n combines with T_n^2 with a high elasticity $\sigma_2 > 1$:

$$v_n = \left[\mu_n^{\frac{1}{\sigma_2}} (T_n^2)^{\frac{\sigma_2-1}{\sigma_2}} + (1 - \mu_n)^{\frac{1}{\sigma_2}} h_n^{\frac{\sigma_2-1}{\sigma_2}} \right]^{\frac{\sigma_2}{\sigma_2-1}}.$$

The task efficiency wage dual is:

$$w_n^i = \left(\sum_{a \in G} \tilde{\kappa}^{ia} (w_n^{ia})^{1-\tilde{\epsilon}} \right)^{\frac{1}{1-\tilde{\epsilon}}}.$$

The cost duals of the two task composites are:

$$w_n^1 = \left(\sum_{i \in I_1} \tilde{\gamma}^i (w_n^i)^{1-\epsilon^1} \right)^{\frac{1}{1-\epsilon^1}}, \quad w_n^2 = \left(\sum_{i \in I_2} \tilde{\gamma}^i (w_n^i)^{1-\epsilon^2} \right)^{\frac{1}{1-\epsilon^2}}.$$

The value-added composite price indices are:

$$p_n^h = \left[\delta_n r_n^{1-\sigma_1} + (1 - \delta_n) (w_n^1)^{1-\sigma_1} \right]^{\frac{1}{1-\sigma_1}},$$

$$p_n^v = \left[\mu_n (w_n^2)^{1-\sigma_2} + (1 - \mu_n) (p_n^h)^{1-\sigma_2} \right]^{\frac{1}{1-\sigma_2}}.$$

The marginal cost is:

$$c_n = B_n (\Xi_n)^\gamma (p_n^v)^{1-\gamma}.$$

Labor market clearing for age group a : the wage bill of age group a equals its total demand from the market sector across both task partitions:

$$w_n^a L_n^a = \sum_{i \in I_1} \gamma_n^{a,i \in I_1} Y_n + \sum_{i \in I_2} \gamma_n^{a,i \in I_2} Y_n,$$

where $\gamma_n^{a,i}$ is the share of age group a 's compensation in output Y_n :

$$\gamma_n^{a,i \in I_1} = \underbrace{(1 - \gamma)}_{\text{VA share}} \cdot \underbrace{\left[(1 - \mu_n) \left(\frac{p_n^h}{p_n^v} \right)^{\sigma_2 - 1} \right]}_{h_n \text{ in } v_n} \cdot \underbrace{\left[(1 - \delta_n) \left(\frac{w_n^1}{p_n^h} \right)^{\sigma_1 - 1} \right]}_{T_n^1 \text{ in } h_n} \cdot \underbrace{\left[\tilde{\gamma}^i \left(\frac{w_n^i}{w_n^1} \right)^{1 - \epsilon^1} \right]}_{t_n^i \text{ in } T_n^1} \cdot \underbrace{\left[\tilde{\kappa}^{ia} \left(\frac{w_n^{ia}}{w_n^i} \right)^{1 - \tilde{\epsilon}^i} \right]}_{L_n^{ia} \text{ in } t_n^i},$$

$$\gamma_n^{a,i \in I_2} = \underbrace{(1 - \gamma)}_{\text{VA share}} \cdot \underbrace{\left[\mu_n \left(\frac{w_n^2}{p_n^v} \right)^{\sigma_2 - 1} \right]}_{T_n^2 \text{ in } v_n} \cdot \underbrace{\left[\tilde{\gamma}^i \left(\frac{w_n^i}{w_n^2} \right)^{1 - \epsilon^2} \right]}_{t_n^i \text{ in } T_n^2} \cdot \underbrace{\left[\tilde{\kappa}^{ia} \left(\frac{w_n^{ia}}{w_n^i} \right)^{1 - \tilde{\epsilon}^i} \right]}_{L_n^{ia} \text{ in } t_n^i}.$$

B.2 Theoretical Mechanisms: Derivatives of Factor Demand

To demonstrate the asymmetric impact of automation (a decline in the rental price of capital r_n) on the two task groups, we differentiate the conditional factor demand functions. From the inner CES structure, the bundle price is strictly increasing in the capital price:

$$\frac{\partial p_n^h}{\partial r_n} > 0.$$

Substitution Effect on Routine Workers (T^2)

Holding output v_n constant, the conditional demand for the substitutable composite is:

$$T_n^2 = v_n \cdot \mu_n \left(\frac{w_n^2}{p_n^v} \right)^{-\sigma_2}.$$

Since T_n^2 and the bundle h_n are substitutes in the outer nest ($\sigma_2 > 1$), the cross-price elasticity is positive:

$$\frac{\partial T_n^2}{\partial r_n} = \frac{\partial T_n^2}{\partial p_n^h} \frac{\partial p_n^h}{\partial r_n} > 0.$$

A fall in the capital price ($\partial r_n < 0$) therefore reduces demand for routine labor ($\partial T_n^2 < 0$).

Complementarity Effect on Non-Routine Cognitive Workers (T^1)

The demand for T_n^1 is nested inside the bundle h_n :

$$T_n^1 = h_n(p_n^h) \frac{\partial p_n^h}{\partial w_n^1}.$$

Substituting CES demand functions gives:

$$T_n^1 \propto (p_n^h)^{\sigma_2 - \sigma_1}.$$

Here σ_2 captures the expansion effect (higher bundle demand raises T^1) and σ_1 captures internal substitution. Differentiating with respect to r_n :

$$\frac{\partial T_n^1}{\partial r_n} \propto (\sigma_2 - \sigma_1) (p_n^h)^{\sigma_2 - \sigma_1 - 1} \frac{\partial p_n^h}{\partial r_n}.$$

Under $\sigma_2 > \sigma_1$:

$$\sigma_2 - \sigma_1 > 0 \implies \frac{\partial T_n^1}{\partial r_n} > 0.$$

A fall in the capital price therefore raises demand for non-routine cognitive labor ($\partial T_n^1 > 0$), confirming the capital-skill complementarity in the market sector.

C Derivation of the Amenity-Adjusted Payoff Decomposition

Derivation of the Amenity-Adjusted Payoff Decomposition In this section, we provide the log-linear decomposition for the amenity-adjusted market payoff. By applying a first-order approximation to the price index change, we decompose the total payoff shift into five additive components. Let U_n^{ia} denote the amenity-adjusted market payoff of a worker of task i and age a in region n . By definition:

$$U_n^{ia} = \frac{w_n^{ia} B_n^a}{P_n^M},$$

where w_n^{ia} is the nominal task wage, B_n^a is the age-specific endogenous amenity, and P_n^M is the market-sector price index. Taking log differences yields:

$$\ln \hat{U}_n^{ia} = \ln \hat{w}_n^{ia} + \ln \hat{B}_n^a - \ln \hat{P}_n^M.$$

Following the log-linearization of the Eaton-Kortum price index around the baseline expenditure shares λ_{nm} , the change in the price index is approximated as:

$$\ln \hat{P}_n^M = \sum_m \lambda_{nm} \left(\ln \hat{c}_m + \ln \hat{\tau}_{nm} - \frac{1}{\iota} \ln \hat{\tau}_m \right),$$

where ι is the trade elasticity. The unit production cost and technology-scale terms are defined

as:

$$\begin{aligned}\ln \hat{c}_m &= \beta \left(\sum_i \Omega_m^i \ln \hat{w}_m^i \right) + (1 - \beta) \ln \hat{P}_m^M, \\ \ln \hat{T}_m &= \gamma A \ln \hat{\Phi}_m.\end{aligned}$$

Substituting these components into the payoff equation yields the final linear decomposition:

$$\begin{aligned}\ln \hat{U}_n^{ia} &= \ln \hat{w}_n^{ia} + \ln \hat{B}_n^a \\ &\quad + \sum_m \lambda_{nm} \left[\frac{\gamma A}{\iota} \ln \hat{\Phi}_m \right] \\ &\quad - \sum_m \lambda_{nm} \beta \left[\sum_i \Omega_m^i \ln \hat{w}_m^i \right] \\ &\quad - \sum_m \lambda_{nm} \left[(1 - \beta) \ln \hat{P}_m^M + \ln \hat{\tau}_{nm} \right].\end{aligned}$$

The terms on the right-hand side correspond to own labor scarcity, endogenous amenities, technical access, task cost bottlenecks, and non-labor input linkages, respectively.